TPT Agent

Algorithm Documentation

Tradier Paper Trading — Full Strategy Reference

Strategy	Cash-Secured Puts (CSP) + LEAPS Deep ITM Calls
Broker	Tradier Sandbox (Paper Trading) — Account VA54665450
Schedule	Daily at 6:35 AM PT (every day)
Data	Tradier real-time APIs + yfinance (earnings only)
Discord	#beta-ai-trades — Summary cards only
Version	Generated June 14, 2026

1. Overview

The TPT Agent is a fully automated options trading bot that runs on Tradier's paper trading sandbox. It executes two parallel options strategies on the same curated stock watchlist: Cash-Secured Puts (CSP Wheel strategy) and LEAPS deep ITM calls (stock replacement). The bot runs once per day at 6:35 AM PT, screens the entire approved stock universe, manages existing positions, and opens new ones based on a rules-based scoring system.

Key design principles: (1) Never trade below the 200-day SMA. (2) Never sell a CSP into an earnings window. (3) Size up only on high-conviction setups (score 5/5). (4) Close winners early — CSPs at 50% premium captured, LEAPS at +10% profit. (5) VIX gates both deployment amount and LEAPS eligibility.

System Architecture

Component	Detail
Bot file	tpt_agent.py
Scheduler	macOS launchd — com.ankushsinghal.tptagent.plist
Trigger time	6:35 AM PT, every day
Broker	Tradier sandbox (paper trading)
Account	VA54665450
Stock universe	Google Sheets CSV — curated approved watchlist
Options data	Tradier markets/options/chains (greeks=true) — real-time
Stock price data	Tradier markets/history + markets/quotes — real-time
VIX	Tradier markets/quotes ($VIX.X) — yfinance fallback
Earnings filter	yfinance only (Tradier has no earnings calendar)
Beta calculation	Computed from Tradier 3-month history vs SPY
Order type	Limit orders at live mid-price (bid+ask)/2
Log file	tpt_agent.log
Discord channel	#beta-ai-trades (hardcoded webhook — no other channel)
Discord output	Run summary + open positions table + CSP summary + LEAPS summary

2. Data Sources

All market data is fetched in real-time via Tradier's REST APIs. yfinance is retained only for earnings dates, which Tradier does not provide.

Data Point	Source	Latency	Used For
Stock OHLCV bars	Tradier /markets/history	Real-time	SMA 20/50/200, Bollinger Bands, RSI, beta
Real-time stock quote	Tradier /markets/quotes	Real-time	Current price, day change %
Options chain (bid/ask/OI)	Tradier /markets/options/chains	Real-time	Strike selection, ARR, OI filter
Implied Volatility (IV)	Tradier greeks.mid_iv	Real-time	IV scoring criterion, BS delta fallback
Options Delta (CSP)	Tradier greeks.delta	Real-time	Delta filter vs VIX-adjusted range
Options Delta (LEAPS)	Black-Scholes computed	Real-time	Always recomputed — Tradier greeks unreliable for 1-2yr options
VIX	Tradier $VIX.X → yfinance fallback	Real-time	Deploy % and LEAPS gate
Beta vs SPY	Computed from Tradier history	Real-time	LEAPS Bollinger Band criterion
Earnings dates	yfinance (only yfinance use)	Daily	Hard filter — skip if earnings in DTE window
Option expirations	Tradier /markets/options/expirations	Real-time	Finding valid DTE windows
Live mid-price at execution	Tradier /markets/quotes	Real-time	Limit price on every order placed

3. Run Pipeline — 9 Phases

Every time the bot fires at 6:35 AM PT it executes nine sequential phases. Each phase depends on the results of the previous one.

1 VIX + Account Snapshot

Fetch VIX, portfolio value, cash, and all open positions from Tradier.

2 Manage Existing Positions

Scan open positions: close CSPs at 50% premium captured, close LEAPS at +10% profit.

3 Refresh Capital After Closes

Re-fetch account after any closes to get updated cash and buying power.

4 Load Tickers + Hard Filters

Pull approved stock list, fetch price history, apply 200 SMA and earnings hard filters.

5 CSP Screening + Scoring

Score each stock on 5 technical signals, find best put contract, rank by score then ARR/Delta ratio.

6 LEAPS Screening

If VIX ≤ 18 or ≥ 21: score stocks on 3 LEAPS criteria, find best deep ITM call contract.

7 Execute LEAPS Trades

Top-ranked first, 1 contract per pick, until 15% portfolio budget exhausted.

8 Execute CSP Trades

Top-scored first, 1 contract per pick, until VIX-adjusted cash budget exhausted.

9 Post Discord Summary

Post run summary, open positions table, CSP summary table, LEAPS summary table.

4. Phase 1 — VIX & Account State

VIX → CSP Delta Range

VIX Level	Regime	Put Delta Range	Rationale
> 20	High	0.10 – 0.20	High vol = fat premium further OTM
15 – 20	Moderate	0.10 – 0.20	Standard range
< 15	Low	0.20 – 0.35	Low vol = move closer to ATM for premium

VIX → Cash Deployment Percentage

VIX Level	Deploy %	Rationale
> 25	80%	Fear = opportunity, deploy aggressively
20 – 25	70%	Elevated vol, still good entry conditions
15 – 20	60%	Moderate vol, standard deployment
12 – 15	40%	Low vol, be selective and conservative
< 12	20%	Very low vol, premiums too thin to deploy much

Budget Calculations

- **CSP Budget** = Cash × VIX deploy %
- **LEAPS Budget** = Portfolio Value × 15% – Current LEAPS Exposure (never exceeds 15% of portfolio total)
- **LEAPS VIX Gate** = Screen and execute LEAPS when **VIX ≤ 18 OR VIX ≥ 21**

> *Rationale for LEAPS VIX gate (≤18 or ≥21):* VIX ≤ 18 = calm market, stocks fairly valued — ideal LEAPS entry. VIX ≥ 21 = fear-driven selloff, stocks oversold — excellent LEAPS entry at depressed prices. Zone 18–21 excluded: moderate stress where IV is neither cheap nor justified.

5. Phase 2 — Position Management (Close Winners)

Before opening any new positions, the bot scans every open position and closes those that have reached their profit target.

CSP Close Rule — Short Puts

Trigger: (Avg Entry Price – Current Price) / Avg Entry Price ≥ 50%

Meaning: We sold the put at $X per share. If it now costs ≤ $X/2 to buy back, 50% of the maximum premium has been captured.

Action: Place a limit BUY TO CLOSE order at the live Tradier mid-price.

LEAPS Close Rule — Long Calls

Trigger: (Current Price – Avg Entry Price) / Avg Entry Price ≥ 10%

Meaning: The LEAPS call we bought is up 10% or more in value.

Action: Place a limit SELL TO CLOSE order at the live Tradier mid-price.

Order Execution Mechanics

- Fetch live bid/ask from Tradier **markets/quotes** endpoint
- Calculate mid-price: **(bid + ask) / 2**
- Place a **limit order at mid-price**, time-in-force = day
- If no live quote available: fallback to last known price

6. Phase 4 — Stock Universe & Hard Filters

The bot loads a curated watchlist from Google Sheets, fetches 310 calendar days of daily OHLCV bars from Tradier for each ticker, computes technical indicators, then applies two hard filters. Any stock failing either filter is completely excluded from both CSP and LEAPS screening.

Technical Indicators Computed

Indicator	Calculation	Used In
SMA 20	Mean of last 20 daily closes	Bollinger Band midline, CSP scoring
SMA 50	Mean of last 50 daily closes	CSP scoring (strike above 50 SMA)
SMA 200	Mean of last 200 daily closes	Hard filter — must be above this
Bollinger Upper	SMA 20 + 2 × std(last 20 closes)	Reference
Bollinger Lower	SMA 20 – 2 × std(last 20 closes)	LEAPS scoring (near lower band)
RSI (14)	Wilder's smoothed 14-period RSI	LEAPS scoring (not overbought)
Day Change %	(current – prev_close) / prev_close	CSP scoring (healthy pullback)
Beta vs SPY	Cov(stock,SPY) / Var(SPY) — 3 months	LEAPS BB criterion (beta-adjusted)

Hard Filters

Filter	Logic	Skip if
200-day SMA	Is stock in long-term uptrend?	Price ≤ SMA 200
Earnings in DTE window	Earnings report coming up?	Earnings date falls within next MAX_DTE (45) days

■ *Earnings data is fetched from yfinance — the only remaining yfinance dependency. Tradier does not provide an earnings calendar.*

7. Phase 5 — CSP Screening & Scoring

Step 1 — RSI Hard Gate (NEW)

If RSI ≥ 65 → SKIP immediately. Overbought stocks have high mean-reversion risk within the 30–45 DTE window. No options lookup is performed.

Step 2 — BB-Based Delta Range (NEW — replaces VIX-global range)

BB Position	Delta Range	Rationale
Price ≤ Lower BB + 3%	0.25 – 0.35 (Aggressive)	Stock oversold → high reversal probability → sell closer to ATM
Between Lower BB and Mid BB	0.15 – 0.25 (Moderate)	Partial pullback → standard entry quality
Price ≥ Mid BB (SMA 20)	0.08 – 0.15 (Conservative)	At/above average → more downside room → stay far OTM

Step 3 — Technical Scoring (max 5 points)

Signal	Points	Condition
Above 50-day SMA	+1	Stock price > SMA 50 (medium-term uptrend)
At/below 20-day SMA	+1	Stock price ≤ SMA 20 / BB midline (short-term pullback)
Healthy pullback today	+1	Day change between –0.5% and –5.0%
RSI < 50	+1	Actively oversold — best entry timing (replaces price<$100)
IV ≥ 40%	+1	Elevated implied volatility — premium is rich

Require final score ≥ MIN_SCORE_TO_TRADE (3) to qualify. Score = 5 → flagged HIGH CONVICTION.

Step 4 — Find Best Put Contract (Tradier Options Chain)

For each expiration date between MIN_DTE (30) and MAX_DTE (45) days:

- Fetch full options chain from Tradier with **greeks=true**
- Filter puts: Open Interest ≥ 100, valid bid/ask quote
- Get delta from **Tradier greeks.delta**; fallback to Black-Scholes if null
- Apply BB-position-based delta filter (per-stock tier: aggressive / moderate / conservative)
- Calculate **ARR = (mid / strike) × (365 / DTE) × 100**

Strike Selection Logic:

If best ARR ≤ MAX_ARR (70%): Pick the contract with the highest ARR in range.
If best ARR > MAX_ARR: Step down to the highest ARR that is still ≤ 70%.
If all contracts exceed ARR cap: Pick the contract with the lowest delta (safest).

Step 5 — Rank and Select Top 5

- Sort qualifying CSPs by the following priority:
- **1. Score DESC** — technical setup quality (primary)

- **2. ARR ÷ Delta DESC** — normalized risk-adjusted return (secondary)
- Rationale: a lower delta + higher ARR means the market pays you MORE for taking LESS
- directional risk. ARR/Delta correctly ranks e.g. delta=0.14 ARR=69% (ratio=493)
- above delta=0.20 ARR=47% (ratio=235). Pure ARR alone would invert this.
- **3. DTE DESC** — more time = more cushion for recovery
- **4. ARR DESC** — final tie-breaker only
- Take top 5

8. Phase 6 — LEAPS Screening

VIX Hard Gate: LEAPS screening runs ONLY when **VIX ≤ 18 OR VIX ≥ 21**. Zone 18–21 is excluded (moderate stress — IV inflated without directional clarity). Outside both zones the entire LEAPS phase is skipped.

Runs on the same stock universe that passed the hard filters in Phase 4.

LEAPS Scoring Criteria (3 points total)

Criterion	Points	Logic	Notes
Earnings safety	+1	Next earnings > 15 days away	Unknown date = allow through with warning
Bollinger Band position	+1	Beta ≤ 2.2: price within 5% of lower BB Beta > 2.2: price at/below SMA 20	Beta-adjusted — high-beta stocks need tighter filter
RSI not overbought	+1	RSI(14) < 70	Wilder's smoothed RSI from Tradier daily bars

- Minimum score to qualify for LEAPS: **2 out of 3**

LEAPS Contract Selection (Tradier Options Chain)

- Expiration window: **365 to 730 days (1–2 years out)**
- Contract type: **Call options only**
- OI filter: skip only if confirmed OI > 0 but < 50 (OI = 0 means data unavailable, allow through)
- Price: (bid+ask)/2 → ask → bid → lastPrice (in priority order)
- Delta: use **Tradier greeks.delta** if available, otherwise compute via **Black-Scholes**
- For BS computation: use Tradier IV if IV ≥ 0.20, else default to **0.45** (LEAPS greeks are unreliable in data feeds)
- Delta filter: **0.70** ≤ **delta** ≤ **0.85** (deep in the money, lowered from 0.80–0.99)
- Selection: pick the contract whose delta is **closest to 0.77 (target)**

LEAPS Ranking

- Sort by: **leaps_score DESC**, then **RSI ASC** (most oversold first)
- Take top 5

9. Phases 7 & 8 — Trade Execution

LEAPS Execution (Phase 7)

Budget: Portfolio Value × 10% (minus current LEAPS exposure already on the books)

Allocation: Top-ranked first. 1 contract per pick. Full remaining budget available for each pick.
Skip a pick only if the remaining budget is too small for its cost — move to the next pick.

For each LEAPS pick (in ranked order):

- If remaining budget < cost_per_contract → skip, try next
- Fetch live mid-price from Tradier quotes
- Place **limit BUY TO OPEN** order at mid-price, qty = 1, time-in-force = day
- Deduct cost from remaining budget and continue to next pick

CSP Execution (Phase 8)

Budget: Cash × VIX deploy %

Collateral: Each CSP requires Strike × $100 in cash as collateral.
Allocation: Top-scored first. 1 contract per pick.

For each CSP pick (in score-ranked order):

- If remaining cash < strike × $100 → skip, try next
- Fetch live mid-price from Tradier quotes
- Place **limit SELL TO OPEN** order at mid-price, qty = 1, time-in-force = day
- Deduct collateral from remaining cash and continue to next pick

Order Format (Tradier API)

All orders are placed as form-encoded POST requests to Tradier:

Parameter	CSP Value	LEAPS Value
class	option	option
symbol	Underlying ticker	Underlying ticker
option_symbol	OCC put symbol	OCC call symbol
side	sell_to_open	buy_to_open
quantity	1	1
type	limit	limit
price	Live Tradier mid	Live Tradier mid
duration	day	day

10. Key Parameters Reference

All parameters are set via environment variables in the launchd plist (com.ankushsinghal.tptagent.plist). The defaults below are the current live values.

CSP Parameters

Parameter	Default	Description
MIN_DTE	30	Minimum days to expiration for CSP puts
MAX_DTE	45	Maximum days to expiration for CSP puts
MIN_ARR	40%	Minimum annualized return on risk to qualify
MAX_ARR	70%	ARR cap — step down to safer strike if exceeded
MIN_OPEN_INTEREST	100	Minimum open interest for liquidity
CSP_RSI_OVERBOUGHT	65	Hard gate — skip stock if RSI ≥ this (overbought)
CSP_DELTA_NEAR_LOWER_MIN/MAX	0.25/0.35	Delta range when price ≤ lower BB + 3% (aggressive)
CSP_DELTA_MID_ZONE_MIN/MAX	0.15/0.25	Delta range when price between lower and mid BB (moderate)
CSP_DELTA_ABOVE_MID_MIN/MAX	0.08/0.15	Delta range when price ≥ mid BB (conservative)
MIN_SCORE_TO_TRADE	3	Minimum score (out of 5) to qualify for execution
SIZE_UP_SCORE	5	Score threshold for HIGH CONVICTION label
SCORE_IV_MIN_PCT	40%	IV threshold for +1 scoring point
SCORE_RSI_OVERSOLD	50	RSI below this earns +1 score point (actively oversold)
SCORE_DOWN_TODAY_MIN_PCT	0.5%	Min pullback today to earn the pullback score point
SCORE_DOWN_TODAY_MAX_PCT	5.0%	Max pullback today (beyond this = potential problem)
CSP_CLOSE_PREMIUM_PCT	50%	Close CSP when this % of premium has been captured
CSP_DTE_EXIT	21	Close CSP at DTE ≤ this with positive/neutral P&L; (gamma gate)
CSP_STOP_LOSS_MULT	2.0×	Close CSP when loss ≥ 2× premium received

LEAPS Parameters

Parameter	Default	Description
LEAPS_MIN_DTE	365 days	Minimum DTE for LEAPS contracts
LEAPS_MAX_DTE	730 days	Maximum DTE for LEAPS contracts (1–2 years)
LEAPS_MIN_DELTA	0.80	Minimum call delta (deep ITM)
LEAPS_MAX_DELTA	0.99	Maximum call delta
LEAPS_TARGET_DELTA	0.85	Target delta — pick contract closest to this
LEAPS_VIX_LOW	15	Lower VIX bound for LEAPS gate
LEAPS_VIX_HIGH	18	Upper VIX bound for LEAPS gate
LEAPS_MIN_SCORE	2	Minimum LEAPS criteria score (out of 3)

Parameter	Default	Description
LEAPS_MIN_OI	50	Minimum OI (only enforced when OI > 0 from API)
LEAPS_MAX_PORTFOLIO_PCT	15%	Hard cap on total LEAPS exposure as % of portfolio (raised from 10%)
LEAPS_RSI_OVERBOUGHT	70	RSI threshold — below this = not overbought (+1 pt)
LEAPS_RSI_PERIOD	14	RSI calculation period (Wilder's smoothed)
LEAPS_HIGH_BETA	2.2	Beta above which stricter BB criterion applies
LEAPS_BB_MAX_DIST_PCT	5%	Max % above lower BB for normal-beta stocks
LEAPS_MIN_EARNINGS_DAYS	15	Minimum days to next earnings for LEAPS safety
LEAPS_CLOSE_PROFIT_PCT	10%	Close LEAPS when unrealized profit reaches this

11. Discord Output

All output goes exclusively to **#beta-ai-trades**. The webhook URL is hardcoded in tpt_agent.py — it cannot be redirected to another channel via configuration. Only summary tables are posted; no individual trade detail cards.

Post #	Content	Always Posted?	Format
1	Run Summary	Yes	Embed: opening balance, VIX, positions closed, positions opened, closing balance
2	Open Positions Table	If positions exist	Code block: symbol, type, qty, entry, current, P&L; $, P&L; %
3	LEAPS Criteria Embed	Yes	Embed: VIX gate status, criteria list, DTE/delta targets
4	LEAPS Summary Table	If VIX in range + picks found	Code block: ticker, score, strike, expiry, cost, delta
5	CSP Summary Table	If picks found	Code block: ticker, score, strike, expiry, premium, ARR, delta

All embeds and tables include [TPT] label to distinguish from the Experiment Bot which posts to the same channel.

12. Full Decision Flow

START 6:35 AM PT trigger

Fetch VIX → determine deploy %; LEAPS gate: VIX ≤ 18 OR ≥ 21

Fetch Tradier account: portfolio value, cash, option buying power

Fetch all open positions

■■■ POSITION MANAGEMENT ■■■

For each open SHORT PUT (CSP):

- Fetch live Tradier mid-price
- If (entry – current) / entry ≥ 50% → BUY TO CLOSE at mid

For each open LONG CALL with DTE > 200 (LEAPS):

- Fetch live Tradier mid-price
- If (current – entry) / entry ≥ 5% → SELL TO CLOSE at mid

Re-fetch account to capture released capital

■■■ STOCK SCREENING ■■■

Load approved ticker list from Google Sheets

For each ticker:

- Fetch 310 days OHLCV from Tradier → compute SMA20/50/200, BB, RSI, beta
- Fetch real-time quote from Tradier
- HARD FILTER 1: skip if price ≤ SMA 200
- HARD FILTER 2: skip if earnings within MAX_DTE days (yfinance)

■■■ CSP SCORING ■■■

For each stock passing hard filters:

- Hard gate: RSI ≥ 65 → skip (overbought)
- BB-based delta range: near lower BB=0.25–0.35 / between BBs=0.15–0.25 / above mid=0.08–0.15
- Score: above 50 SMA (+1), below mid BB (+1), pullback 0.5–5% (+1), RSI<50 (+1), IV≥40% (+1)
- Fetch put chain from Tradier (greeks=true)
- Filter puts: OI≥100, delta in BB-tier range, ARR 40–70%
- If final score < 3: skip

Sort CSPs: score DESC → ARR/Delta DESC → DTE DESC → ARR DESC → take top 5

■■■ LEAPS SCORING (only if VIX ≤ 18 OR ≥ 21) ■■■

For each stock passing hard filters:

- Criterion 1: earnings > 15 days away (+1)
- Criterion 2: near lower Bollinger Band or beta-adjusted midline (+1)
- Criterion 3: RSI(14) < 70 (+1)
- If score < 2: skip
- Fetch call chain from Tradier (DTE 365–730, greeks=true)
- Filter calls: delta 0.70–0.85; pick closest to delta 0.77

Sort LEAPS by score DESC, RSI ASC → take top 5

■■■ EXECUTION ■■■

LEAPS: iterate top-ranked first, 1 contract each, until 15% budget exhausted
CSP: iterate top-scored first, 1 contract each, until VIX-deploy cash exhausted
All orders: limit at live Tradier mid-price, time-in-force = day
■■■ DISCORD ■■■
Post run summary, open positions, CSP table, LEAPS table to #beta-ai-trades
END

** The experiment_bot.py (Alpaca) runs an identical strategy independently. Trade picks are generated separately for each broker — they are not shared. This allows side-by-side comparison of broker execution quality.*